

Quantum-Classical Hybrid Architectures for Adversarial Citation Integrity Verification

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Abstract

We introduce a novel hybrid architecture for automated manuscript evaluation and citation integrity verification. By combining dual-agent adversarial consensus protocols with multi-source retrieval across arXiv and PubMed Central, our engine isolates claim overreach and unstated methodological caveats. Experimental evaluation across 1,200 peer-reviewed manuscripts demonstrates 94.6% agreement with human expert benchmarks while reducing manual audit latency by 85%. Furthermore, we present the Two Pens stroke resolution algorithm for visual passage comparison.

1. Primary Manuscript Claims

Our empirical findings establish four core principles governing automated citation verification:

[1] Claim 1 (Multimodal Transfer): Pre-trained visual-language representations achieve 76.2% zero-shot top-1 accuracy across diverse computer vision benchmarks [1].

[2] Claim 2 (Adversarial Red-Teaming): Dual-agent adversarial red-teaming isolates subtle claim overreach and unstated methodological caveats in scientific literature [2].

[3] Claim 3 (Attention Complexity): Quantum-assisted tensor decomposition reduces self-attention layer computational complexity from quadratic $O(N^2)$ to logarithmic $O(N \log N)$ [3].

[4] Claim 4 (Cost Auditability): Itemized token cost accounting provides transparent real-time auditability across heterogeneous LLM pipeline executions [4].

2. Cited References & Literature

[1] Radford et al., "Learning Transferable Visual Models From Natural Language Supervision", ICML 2021.

[2] Vaswani et al., "Attention Is All You Need", NeurIPS 2017.

[3] Brown et al., "Language Models are Few-Shot Learners", NeurIPS 2020.

[4] Kaplan et al., "Scaling Laws for Neural Language Models", arXiv 2020.